



State space model :

The general form of a state space model is as follows :

$$\left. \begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \right\} \begin{array}{l} \text{Continuous} \\ \text{time} \end{array}$$

$$\left. \begin{aligned} x[k+1] &= Ax[k] + Bu[k] \\ y[k] &= Cx[k] + Du[k] \end{aligned} \right\} \begin{array}{l} \text{discrete} \\ \text{time} \end{array}$$

x is a vector and its elements are called the "states" of the system.

A is a matrix with dimension $n \times n$. It is referred to as the "system matrix" or "state transition matrix"

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & & \vdots \\ A_{n1} & A_{n2} & & A_{nn} \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix}$$

B = input matrix
(dimension $n \times 1$)
(column vector)

C = output matrix (row vector)
(dimension is $1 \times n$ usually)

' y ' is called the output of the system.

D is a scalar and it is called the "direct transmission term."

This approach of notifying a system is also called "state variable form".

Let's see an example. Let's consider the differential equation of a DC

motor.

$$J_m \ddot{\theta}_m + b \dot{\theta}_m = k_t i_a$$

$$L_a \frac{di_a}{dt} + R_a i_a = V_a - k_e \dot{\theta}_m$$

This is a coupled differential equation. The two equations are related through the current i_a and speed $\dot{\theta}_m$. Let's form our state vector x .

$$x = \begin{bmatrix} \theta_m \\ \dot{\theta}_m \\ i_a \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$x_1 = \theta_m$$

$$\Rightarrow \dot{x}_1 = \dot{\theta}_m = x_2$$

$$x_2 = \dot{\theta}_m$$

$$\Rightarrow \dot{x}_2 = \ddot{\theta}_m = \frac{k_t}{J_m} i_a - \frac{b}{J_m} \dot{\theta}_m$$

$$\dot{x}_2 = \frac{k_t}{J_m} x_3 - \frac{b}{J_m} x_2$$

$$x_3 = i_a$$

$$\Rightarrow \dot{x}_3 = \frac{di_a}{dt} = \frac{V_a}{L_a} - \frac{R_a}{L_a} i_a - \frac{k_e}{L_a} \dot{\theta}_m$$

$$\dot{x}_3 = \frac{V_a}{L_a} - \frac{k_e}{L_a} x_2 - \frac{R_a}{L_a} x_3$$

let's write it down in the matrix form now.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{b}{J_m} & \frac{k_t}{J_m} \\ 0 & -\frac{k_e}{L_a} & -\frac{R_a}{L_a} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_a} \end{bmatrix} u \quad \begin{matrix} \uparrow \\ (u=V_a) \end{matrix}$$

$$\dot{x} = Ax + Bu$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 0 \cdot u$$

$$y = Cx + Du$$

This is the state space representation of a DC motor.

Note: The eigenvalues of the matrix A are the poles of the transfer function. In order for a system

to be stable, the eigenvalues has to be negative.